The Color Gives It Away

A cow-derived oxygen carrier looks like the perfect secret for an endurance athlete, and it may be the one doping agent that cannot hide in a tube of plasma

Every few years the idea resurfaces in locker-room folklore and in the margins of sports journalism: somewhere there is a bag of red liquid made from cattle, it carries oxygen, it needs no matching and no refrigerator, and an athlete could in principle use it to borrow a few percent of performance from a patient in an emergency room. The idea is attractive precisely because it sounds like a loophole in the physics of competition. This essay takes the idea seriously and then examines what the public record actually supports. The surprising conclusion, from the documents I could read, is that a BHOC, a Biologic Hemoglobin Oxygen Carrier, is a poor candidate for the perfect crime, because the same property that makes it useful in medicine, the free hemoglobin dissolved in plasma, is a signature that laboratories began learning to read before the product was ever approved for a person. That history says as much about the products as it does about sport, and it contains lessons for anyone trying to bring an oxygen carrier to patients without letting it acquire a reputation as a cheat's drug.

The temptation on paper

The case for why an athlete would want such a product is laid out plainly in a research project hosted on the World Anti-Doping Agency website. The project, coded 01B01RK, was approved in 2001, led by R. Kazlauskas at the Australian Sports Drug Testing Laboratory, and is listed as completed. Its summary observes that cell-free hemoglobin carriers could appeal to athletes for three reasons. Unlike a transfusion, they carry no risk of blood-type mismatch. They do not need refrigeration, and the summary says they have a shelf life of up to two years at room temperature. And they act almost immediately, in contrast to erythropoietin, which has to be taken for weeks before the red cell count rises. The same page notes that erythropoietin and transfusion were already detectable, including by the combined blood and urine test used at the Sydney Olympics in 2000, which is the context in which a chemist would worry about the next thing an unscrupulous team might reach for.

That list speaks of convenience and speed, which are logistical virtues, and says nothing about whether the physiology rewards the athlete. A drug that is easy to use is still a bad bet if it does not make you faster.

What the athlete would not get

I want to separate what I can cite from what I am reasoning. The cited part is short. A bovine hemoglobin purification patent from the 1990s that I read for an earlier essay reports a circulatory half-life of about twenty-four hours for its product and a hemoglobin concentration of about ten grams per deciliter. A day is a short time in the logic of training, where adaptations are built over weeks, and it is a long time in the logic of a single race. The product does not make the body produce more red cells; it adds a second, dissolved oxygen compartment to the plasma that disappears on a schedule, an ephemeral boost at best.

The reasoning part is mine, and it should be treated as an argument, not a finding. Endurance performance is limited by several links in a chain: the lungs loading blood, the heart pumping it, the vessels distributing it, the muscle extracting it, and the mitochondria using it. Adding hemoglobin to plasma addresses the carrying capacity of the blood, and that is only one link. The cell-free molecule also has well-documented side effects in patients, including raised blood pressure and vasoconstriction, which the same WADA-hosted summary alludes to when it says some approaches were limited by such problems. A vasoconstricting agent given to a person who needs maximal blood flow to working muscle is a strange tool. I do not know of a controlled study of exercise performance with a cow-derived carrier in healthy athletes, and my search for one came up empty. It may exist behind a paywall I could not open, or it may never have been done, because an ethics committee asked to approve infusing a healthy volunteer with an agent whose class carried the safety signal of the early 2000s would have had hard questions.

What I can say is that the people with the strongest motive to claim a performance benefit, those writing about the threat, wrote about convenience and not about proof of effect. The enthusiasm for the idea in the doping imagination has, as far as the documents I read show, run well ahead of the evidence that it works.

A pigment that cannot hide

Now the part that surprised me. The detection strategy that the Australian project describes begins with something almost comically simple: looking at the plasma. Hemoglobin that has left the red cell stains the liquid it floats in. The project summary lists visual inspection of plasma for discoloration as the screening step and reports that it detects the carrier at under one percent in plasma. Anyone who has seen a centrifuged tube of blood knows the normal result, a pale straw-colored layer on top of a dark red pellet. A tube that comes out pink or red-brown above the pellet is telling the technician that hemoglobin is in the wrong place. That can happen innocently, from damaged cells in the tube, and the protocols have to distinguish the two, but it means the first sieve costs almost nothing.

The confirmatory methods are more elaborate and are, in my reading, the real intellectual contribution of that decade. The project describes three. The first is size-exclusion chromatography with a photodiode array detector, which sorts proteins by size and identifies the carrier by how long it takes to emerge and by the spectrum of the light it absorbs; it detected a one percent spike. The second is native gel electrophoresis with a luminol chemiluminescence readout, which separates proteins by size and shape and makes hemoglobin glow; the summary states that two products, Hemopure and Oxyglobin, were detectable at one percent. The third is the most telling for a bovine product: enzymatic digestion followed by mass spectrometry, which chops the protein into fragments and identified peaks that came from bovine, not human, hemoglobin. In other words, the laboratory can say not only that foreign hemoglobin is present but that it came from a cow.

The summary adds that these methods were validated on real plasma from a person who had received Hemopure, and that they should detect any chemically cross-linked carrier with a much higher average molecular weight than human hemoglobin, which the authors say covers all of the products in advanced development at the time. The molecular weight range they

quote for Hemopure is 130 to 500 kilodaltons, roughly two to eight times the weight of a single human hemoglobin tetramer, which is about 64 kilodaltons.

A separate group working with a French laboratory published a complementary method in Clinical Chemistry in April 2004. Varlet-Marie and colleagues describe a size-exclusion technique for plasma or serum that separates the cross-linked polymers from human hemoglobin fragments; in the abstract I read, human hemoglobin dimers appear at a peak around fifty-four minutes, well apart from the peaks of the carrier. They tested it in vivo in twelve healthy male volunteers who received Hemopure and report that the serum profile stayed consistent for at least forty-eight hours. They conclude that the method, used together with an electrophoretic screen, provides evidence of the carrier's presence and can meet the criteria needed for anti-doping control. The abstract does not report detection limits, and it notes that products could be told apart by their profiles mainly at high concentrations. I have not read the full paper.

The laboratory as unplanned archive

A pragmatic reader might stop at "so it is detectable." I think the more interesting fact is what the detection literature has quietly become. To build a test for a product, a laboratory has to characterize it: its size distribution, its spectrum, its protein fragments, its behavior in plasma over hours. Those characterizations, produced by people with no commercial interest in the manufacturer and every interest in finding fault, are some of the only independent measurements of these products in the open literature. The manufacturers' own data are mostly in regulatory filings and press releases. The anti-doping chemists published chromatograms.

A 2022 review in Frontiers in Sports and Active Living, which is mainly about detecting transfusions from donors, lists the genealogy of this work in a single sentence: methodology papers by Goebel and colleagues in 2005, by Simitsek and colleagues in 2007, a fast screening method by Donati and Botrè in 2010, a capillary electrophoresis approach by Staub and colleagues in 2010, and a 2011 pre-confirmation method by Donati and colleagues that distinguishes the body's own hemoglobin from the synthetic kind. I could open only the review, not the papers it cites, so I know those studies by title and year and I do not describe their findings. The review itself reports no misuse cases and names no products, a silence I will return to.

Consider what this means for a developer. A company bringing a bovine carrier to patients faces a regulator who wants to know the lot is consistent. An outside laboratory that can separate the polymer fractions, fingerprint the bovine peptides, and follow the serum profile over forty-eight hours has built, for free and in public, a rough analytical dossier. The molecular weight range of 130 to 500 kilodaltons that I quoted is, in effect, a statement about how heterogeneous the product is, and a patient advocate who wants to know whether a given formulation is a tight distribution or a broad smear could do worse than read the doping literature.

The lineage has not stopped. At the 24th International Conference of Racing Analysts and Veterinarians in 2026, an oral presentation described a mass spectrometry method for detecting M101, the oxygen carrier derived from the marine lugworm, in horse plasma. The

abstract gives the molecular weight as 3,600 kilodaltons, says the product is sold by its French developer under a commercial name, and states that it could be misused by human and equine athletes. In spiked horse plasma the method detected the carrier at about one microgram per milliliter. For comparison, the abstract says an earlier electrophoretic method detected the same molecule in spiked human plasma at about fifty micrograms per milliliter, so the new approach is roughly fifty times more sensitive in that comparison. In mice, the carrier was detectable for up to four hours after an intravenous dose of 200 milligrams per kilogram. These are studies in spiked samples and mice, and they say nothing about what happens in a racehorse after a plausible dose; they do show that the detection community is working on worm-derived molecules before any evidence of their abuse has appeared in public. A new class of carrier has barely reached the market, and the testers are already ahead of it.

What I could not find

Honest reporting here requires a list of the things I looked for and did not find, because the absence is informative and because the folklore tends to fill such gaps with confident detail.

I found no documented, adjudicated case of an athlete sanctioned for using a cow-derived carrier. The 2022 review mentions the carriers among the forms of blood doping that accredited laboratories have developed tests for, and it reports no cases. That says little, since the review was not about this subject. I also encountered, in a secondhand encyclopedia mirror, a reference to a public accusation in July 2007 that a leading cyclist had tried to obtain Hemopure. I could not trace the primary reporting, the accusation was made by a rival and not by an investigating body, and I will not name the person. I mention it only to say that the rumor mill exists and that I could not convert it into evidence.

I also could not find how the rules treat a legitimate patient who is an athlete. The Prohibited List bars blood doping at all times, and the United States Anti-Doping Agency’s public FAQ states that all forms of blood doping are prohibited at all times; its page says nothing about oxygen carriers specifically, and it directs readers with product-specific questions to a drug reference service. A competitor with a hemorrhage or a cyclist who refuses transfusion on grounds of conscience after a crash might be treated with a carrier under emergency access. Whether a therapeutic use exemption could apply, and how fast, is a question I could not answer from the documents I found. I raise it because the answer matters for exactly the patients that a conscientious program wants to serve, and because the rules were written for a world in which these products were hypothetical.

Finally, I found no study of whether the carrier improves exercise performance in healthy people. If someone has done that work, I would like to read it. If no one has, then the detection effort of two decades has been aimed at a threat whose payoff was never measured, which is a defensible precaution and a curious one.

Horses and dogs

The strangest corner of this subject is the one that does not involve people. In animal racing, the regulatory posture is blunt. A 2008 press release from the New Jersey Attorney General’s office announced that six harness racehorses had tested positive for human erythropoietin under the state’s expanded testing, were declared ineligible, and that violations carried a ten-

year license suspension and a fine of fifty thousand dollars. The release says nothing about oxygen carriers; I cite it because it shows how seriously a racing jurisdiction treats blood-boosting agents, and the 2026 racing-chemists' abstract shows that the same community regards the worm-derived carrier as a live concern.

The irony sits in the veterinary clinic. A bovine carrier was licensed for dogs, and the encyclopedia mirror I read says its original developer, Biopure, filed for bankruptcy protection in 2009 and that its assets were bought by another company. A product with a legitimate veterinary indication, an approval in South Africa for people in 2001 according to the same mirror, and a plausible illicit audience is a regulatory puzzle: how do you keep a medicine on the shelf for the animal that needs it while keeping it away from the stable that does not? For a bovine carrier the distribution chain is short and the buyers few, which makes lot-level tracking feasible in a way it is not for a pill.

What a patient program should do about it

I would rather end with practical commitments than with a moral. A foundation or developer working toward human access to a BHOC can reduce the risk that the product acquires a cheat's reputation, and doing so costs little.

Share reference material with the anti-doping laboratories early. The Australian project validated its methods against plasma from a real recipient of Hemopure, which means someone in the chain of clinical research cooperated with the testers. That cooperation should be routine, not a happy accident, and it should extend to any new formulation, including a polymerized, encapsulated, or worm-derived one. A laboratory that has the standard before the product is widely available can publish a test before a scandal.

Keep distribution traceable by lot, from the manufacturer to the bedside, and record who received which lot. In a clinical setting this is already good practice for a biologic. Here it has a second use: if a sample ever tests positive, the chain of custody can show whether the source was diversion, a clinical administration to a competing athlete, or a mislabeled specimen.

Say plainly in patient materials that the product is detectable and why. The visible discoloration of plasma that I described is a clinical nuisance for hospital laboratories, because it can disturb tests that read color; it is also the reason the product is a poor tool for deception. Both facts belong in the same paragraph.

Be circumspect about marketing language. Phrases like "boosts oxygen delivery" or "enhances performance" drift easily from a hemorrhage setting into a stadium. A product described as a bridge for a bleeding patient who cannot receive blood is describing a medicine. A product described as a performance enhancer has changed category in the public mind, regardless of intent.

Finally, say what is not known. An advocate who claims the product is both dangerous to athletes and useless to them has said two things that may not both be true.

The deeper lesson

A cell-free carrier is, to a laboratory, a foreign object in a place it does not belong: free in the plasma, large, colored, and bovine in its amino acid sequence. Medicine treated those properties as obstacles to engineer around; sport's chemists treated them as handles. The pigment that stains a patient's plasma, interferes with a hospital's tests, and complicates a clinician's day is the same pigment that makes the product nearly impossible to use unseen. A technology that wears its identity so visibly is, in at least one narrow sense, an honest one.

Further reading

This essay draws on a research project summary hosted on the World Anti-Doping Agency website, led by R. Kazlauskas at the Australian Sports Drug Testing Laboratory; the abstract of a 2004 Clinical Chemistry paper by Varlet-Marie and colleagues; a 2022 review in Frontiers in Sports and Active Living on detecting donor transfusions; an abstract from the 2026 International Conference of Racing Analysts and Veterinarians on detecting the lugworm-derived carrier in horse plasma; the public blood doping FAQ of the United States Anti-Doping Agency; a 2008 New Jersey Attorney General press release on equine erythropoietin; and a bovine hemoglobin patent for the half-life and concentration figures. I read abstracts and summaries, not the full papers, and said so where it matters. Corporate history and the South African approval date come from a secondhand encyclopedia mirror and should be checked against primary sources.